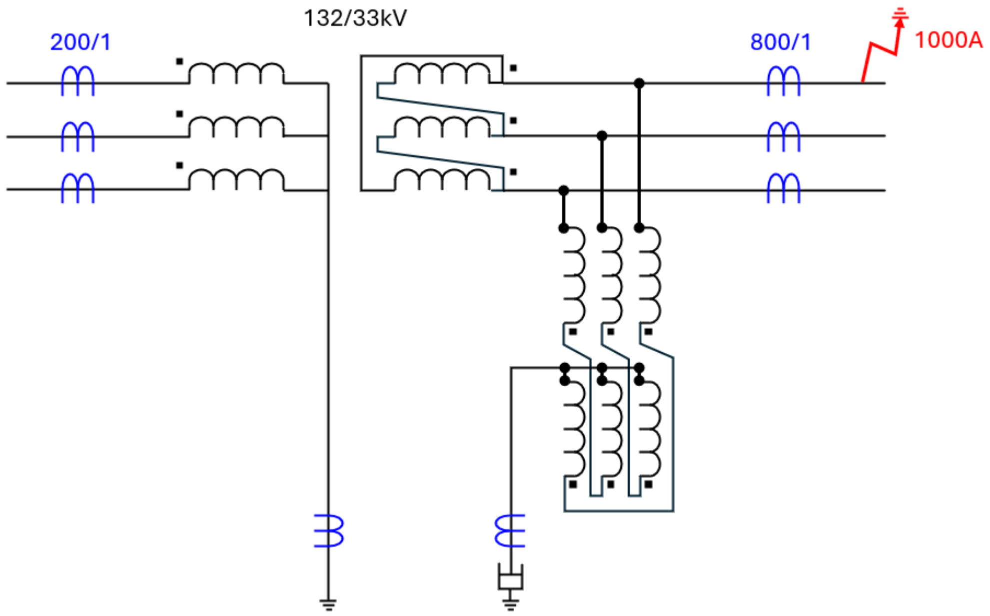
ELEC4142 Power System Protection and Switchgear
Assignment – Transformer Protection
Answer **TWO** questions only. (Total 40 marks)

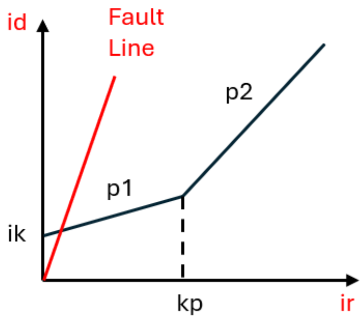
Note – If the question does not provide a tap range, consider the calculation in nominal tap.

1. **[Delta-wye Transformer with Earthing Transformer]**
   a) The following is a 80MVA 132/33kV star-delta transformer with Zigzag transformer grounded at delta side.
      (i) Complete the CT wiring to provide a proper bias differential and restricted earth fault protection such that the design is through fault stable. [4 marks]
      [Hint: At LV side, the ICT should be a star-star, with a tertiary delta-winding to filter the zero sequence current]
      (ii) Determine a proper ICT ratio and indicate the current flow (both primary and secondary) in case there is a L1-E fault with fault current of 1000A. [8 marks]



   b) (i) Is it possible to have an earth fault at delta side in case there is no zigzag transformer at delta side? How to provide a sensitive earth fault protection for the delta winding?
      (ii) Is it possible to remain stable in case the tertiary delta-winding is not provided but there is still a zigzag transformer? What is the trade-off?
      (iii) In case the protection scheme is replaced with a digital bias-differential relay, what should be the LV ICT winding setting to be used to maintain stability and why?
      [8 marks]

2. **[Setting Calculation and On Load Stability Test]**
   Given a 35MVA 132/11kV YNyn0 transformer with impedance 26.38% and tap range +10.5% to -24%. The HV line CT is in 1/400T, Class X; while the LV line CT is in 1/2000T, Class X.
   a) i) Determine the HV and LV ICT ratio to achieve unit load at relay side.
      ii) Determine the largest differential current at any taps with maximum CT error of $\pm$5%.
      iii) Determine the bias differential setting for the transformer.
      iv) During the on-load stability test, the HV load current is 51A, while the LV load current is 630A. At the step of "HV All CTs Normal, LV A-ph CT s/c", the relay trips. What is the operating element? Can you confirm "test result satisfactory"? [10 marks]



b) Determine the operating current with phase-to-phase and phase-to-earth configuration to activate bias differential element. [2 marks]

c) Discuss the difference in restricted earth fault (REF) design and the corresponding challenges in E&M relays and digital relay. [4 marks]

d) Discuss the reasons why directional overcurrent (DOC, 67) element is installed in some transformer LV terminals. What are the possible reasons for the instability of the directional element and how to avoid such instabilities? [4 marks]

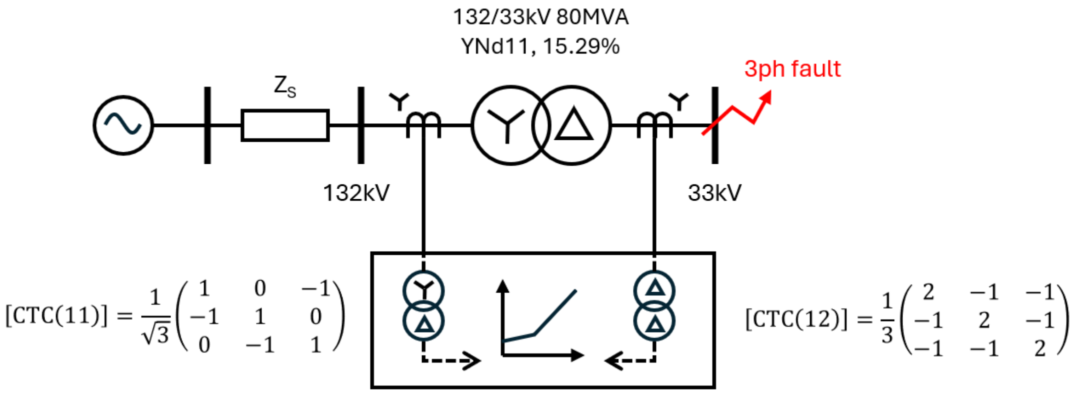
3. **[Star-delta transformer, Fault Level Calculation and Rotation Matrix]**

An 80MVA, 132/33kV, star/delta (YNd11) transformer with percentage impedance 15.29% is earthed solidly on the star side and is supplied with the maximum fault level 6000MVA at 132kV side. Biased differential (87T) protection is used to provide protection for the transformer.

a) Illustrate, with a three-phase wiring diagram, how bias differential protection scheme can be used for the protection of the power transformer. [4 marks]

b) Design a suitable CT and ICT ratios and determine the fault current distribution in the power transformer, CT and ICT secondary, and relays for a phase-to-phase external through fault with fault current of 7000A on the 33kV side. [8 marks]

c) Suppose the transformer protection is replaced with a digital relay, and the ICT has been removed. Determine, under maximum through fault conditions, the differential current and hence if an overcurrent differential fault detector of 0.3A is operative, given that the ICT setting in HV side and LV side are set as YNd11 with [CTC(11)] and Dd0 with [CTC(12)]. Assume the ICT setting has a proper ratio to minimize the differential current at nominal-tap full load condition. [6 marks]



$$[\mathrm{CTC}(11)] = \frac{1}{\sqrt{3}}\begin{pmatrix} 1 & 0 & -1 \\ -1 & 1 & 0 \\ 0 & -1 & 1 \end{pmatrix} \qquad [\mathrm{CTC}(12)] = \frac{1}{3}\begin{pmatrix} 2 & -1 & -1 \\ -1 & 2 & -1 \\ -1 & -1 & 2 \end{pmatrix}$$

d) Discuss if harmonic blocking is useful and meaningful method to maintain the stability of restricted earth fault element in digital relays. [2 marks]

THE END